

Text Classification using Naive Bayes with Sentiment Analysis

1. Introduction

This project focuses on classifying blog posts into categories using the Naive Bayes algorithm and performing sentiment analysis to understand the emotional tone of the content.

2. Dataset

- File: blogs_categories.csv / blogs (3).csv
- Columns:
 - **Data** → Blog text
 - **Labels** → Category

3. Data Preprocessing

Steps:

- Lowercasing text
- Removing punctuation
- Tokenization
- Stopword removal
- Lemmatization

4. Feature Extraction

- Used **TF-IDF Vectorization**

- Converts text into numerical format
- Limited to top 5000 features

5. Model Implementation

- Algorithm: **Multinomial Naive Bayes**
- Split: 80% training, 20% testing

6. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

7. Sentiment Analysis

- Used **VADER Sentiment Analyzer**
- Categories:
 - Positive
 - Negative
 - Neutral

8. Results

- Naive Bayes performs well on text classification
- Sentiment distribution shows variation across categories
- Some categories contain more positive/negative tone

9. Conclusion

- Successfully built a text classification model
- Implemented full NLP pipeline
- Sentiment analysis provides additional insights

10. Key Learnings

- Importance of text preprocessing
- TF-IDF improves model performance
- Naive Bayes works well for text data
- Sentiment analysis adds deeper understanding